Blocco 3

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For the next two problems, suppose that $\mathcal{F}$ is a nonprincipal, k-complete filter on k. Define the dual ideal $I = \{A \subseteq k : k \setminus A \in \mathcal{F}\}$. We say that $A \subset k$ is a positive set if $A \notin I$ (equivalently, for all $C \in \mathcal{F}$, $C \cap A \neq \emptyset$). For a positive set A we say that $f : A \to k$ is regressive if for all $\alpha \in A$, $f(\alpha) < \alpha$.

Esercizio 1. Suppose that for all $A \subseteq k$, $A \notin I$ and regressive function $f : A \to k$, f is constant on a positive subset of A. Show that $\mathcal{F}$ is closed under diagonal intersections.

Proof. Let $C_i \in \mathcal{F}$ for $i < k$ and let $D = \triangle_i C_i$. By contradiction, assume $D \notin \mathcal{F}$. Then $D^c \notin I$, which means that D^c is positive. Consider $f : D^c \to k, \alpha \mapsto \min\{i < \alpha : \alpha \notin C_i\}$.

- f well defined: $\alpha \in D^c \Rightarrow \exists i < \alpha \ \big(\alpha \notin C_i\big)$, so we have that for all $\alpha \in D^c$, $\{i < \alpha : \alpha \notin C_i\} \neq \emptyset$. It has minimum because $\subseteq ON$.
- f regressive: by definition

Let $P \subseteq D^c$ positive such that $f \upharpoonright P = cost_\beta$. Now observe that $P \cap C_\beta = \emptyset$: in fact $p \in P \Rightarrow f(p) = \beta \Rightarrow p \notin C_\beta$. Thus we reached a contradiction because P is positive, $C_\beta \in \mathcal{F}$ but $P \cap \mathcal{F} = \emptyset$.

□

Esercizio 2. Suppose that $\mathcal{F}$ is closed under diagonal intersections. Show that for all $A \subseteq k, A \notin I$ and regressive $f : A \to k$, f is constant on a positive subset of A.

Proof. For all $i < k$, let $N_i = f^{-1}\{i\}$. By contradiction, assume for all $i < k$, N_i is negative (meaning not positive). Then for all $i < k$, let $C_i \in \mathcal{F}$ such that $C_i \cap N_i = \emptyset$. Consider $C = \triangle_i C_i$. Then $C \in \mathcal{F}$. Since A is positive, we have $A \cap C \neq \emptyset$. Let $\alpha \in A \cap C$. Since $\alpha \in C$ we have that $\forall i < \alpha\big(\alpha \in C_i\big)$. This means that $\forall i < \alpha\big(f(\alpha) \neq i\big)$. Equivalently, $f(\alpha) \geq \alpha$, which is a contradiction since f is regressive.

□

For the next two problems let k be a measurable cardinal with normal measure U and let P be the Prickry forcing at k.

Esercizio 3. Suppose that G is P-generic over the ground model V. Let $\vec{\alpha} = \bigcup_{(s,A)\in G} s$ and let $G_{\vec{\alpha}} = \big\{(s, A) : \exists n\big(\vec{\alpha} \upharpoonright n = s, \forall k \geq n \alpha_k \in A\big)\big\}$. Prove that $G = G_{\vec{\alpha}}$.

Proof.

- $G \subseteq G_{\vec{\alpha}}$: Let $(s, A) \in G$. Then by definition of $\vec{\alpha}$ we have that s must be an initial segment of $\vec{\alpha}$. By contradiction, assume $\{n > |s| : \vec{\alpha}(n) \notin A\} \neq \emptyset$. Let m be minimum of the above set. Let $(t, B) \in G$ such that $|t| \geq m$. Such an element exists by genericity of G. Since $(s, A), (t, B)$ must be compatible we have that $t(n) \in A$ for all $n > |s|$. In particular $\vec{\alpha}(m) = t(m) \in A$, contradiction.

- $G_{\vec{\alpha}} \subseteq G$: Let $(s, A) \in G_{\vec{\alpha}}$. Consider $D = \{(t, B) : |t| \geq |s|, B \subseteq A\}$. D is dense because if we take $(z, C) \in P$ (if necessary) we can extend z with the least elements of C until we reach length $|s|$. Let's say we concatenated $\bar{c}$. Then $(z, C) \geq (z \frown \bar{c}, C \setminus \bar{c}) \geq (z \frown \bar{c}, (C \setminus \bar{c}) \cap A) \in D$. Since G generic and D dense, we have $G \cap D \neq \emptyset$. Let $(t, B) \in G \cap D$. From $(t, B) \in G$ we get t initial segment of $\vec{\alpha}$. Combined with $(t, B) \in D$ we deduce t extends s with elements in A. Finally from $(t, B) \in D$ we deduce $B \subseteq A$ and conclude $(t, B) \leq (s, A)$. Since $(t, B) \in G$ we have $(s, A) \in G$.

□

All the notation for the following exercise is intended as the one used in the lessons unless otherwise specified. So for example $j : V \to M$ is elementary embedding, M is the ultraproduct, $k^M = [Id]$, $M^k \subseteq M$, $M \subseteq V$.

Esercizio 4. Suppose that in the ground model V, $2^k = k^{++}$ and G is P-generic over V. Suppose that $\langle k_n : n < \omega \rangle$ is the Prikry sequence added by G. Show that for all large n, k_n is a cardinal and $2^{k_n} = k_n^{++}$.

Proof. Consider $B = \{\lambda \in k : \lambda \text{ is a cardinal and } 2^\lambda = \lambda^{++}\}$. We claim that $B \in U$. This is equivalent to $M \models 2^k = k^{++}$ by Los Theorem.

- $(k^+)^M = (k^+)^V$: In fact, since $M \subseteq V$ we have $(k^+)^M \leq (k^+)^V$. If by contradiction $(k^+)^M < (k^+)^V$, then there would exist $f \in V, f : k \to (k^+)^M$ surjective. Since $M^k \subseteq M$ we have $f \in M$, so $(k = k^+)^M$, a contradiction. In particular from now on we simply write λ to denote $(k^+)^V, (k^+)^M$, since they are the same.

- $(2^k)^M = (2^k)^V$: Since $M \subseteq V$ we have $(2^k)^M \subseteq (2^k)^V$. Since $M^k \subseteq M$ the other inclusion holds as well.

- $(k^{++})^M = (k^{++})^V$: I don't know if it is true. In any case I'd like to receive the solution of the exercise.

Now since in V it is true that $2^k = k^{++}$ and all cardinals in the equation are the same in M, we deduce that the equation holds in M as well.

From $B \in U$ we deduce that $D = \{(s, A) : A \subseteq B\}$ is dense. Then $G \cap D \neq \emptyset$. Let $(s, A) \in G \cap D$. Then for all $n > |s|$ $\big(k_n \in A\big)$, which implies for all $n > |s|$ $\big(k_n \text{ is a cardinal and } 2^{k_n} = k_n^{++}\big)$.

□